

Scientific Computation

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Lecture 30 - Stability and Error of ODEs

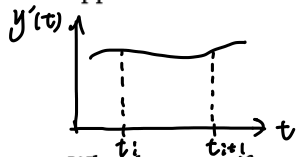
Lecture Outline

- Interpretation as numerical integration
- Stability
- Error

Rectangle Approximations for Numerical Integration

$$\int_{t_1}^{t_2} f(t) dt$$

Recall the problem of numerical integration the interval $[t, t+h]$, but this time applied to functions of the form $f(t, y(t))$:

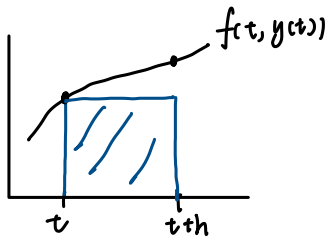


$$I_f = \int_t^{t+h} f(t, y(t)) dt.$$

$$y'(t) = f(t, y(t))$$

$$\int_a^b y'(t) dt = y(b) - y(a)$$

What happens if we approximate this integral using a **left rectangle approximation**? We get something that looks just like the increment for each step of the forward Euler method:



$$I_f \approx h \cdot f(t, y(t))$$

$$y(t_{i+1}) = y(t_i) + h \cdot f(t_i, y(t_i))$$

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Forward Euler's Method Emerges

The forward Euler method is therefore just the repeated application of the left rectangle numerical integration rule on each subinterval $[t_i, t_{i+1}]$ for our IVP

$$y'(t) = f(t, y(t)), \quad y(t_0) = c.$$

That is, we perform the approximate integration

$$y(t_i) = \int_{t_0}^{t_i} y'(t) dt \approx y(t_0) + \sum_{k=1}^i h \cdot f(t_k, y(t_k)).$$

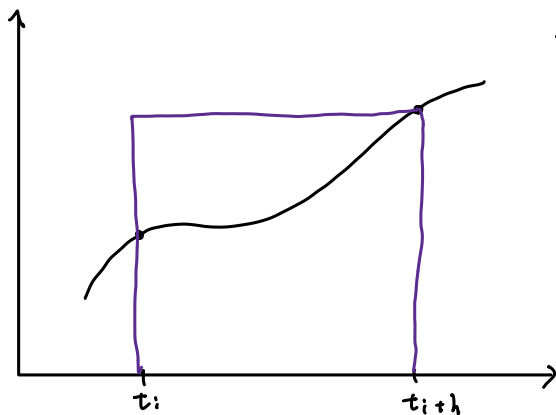
In essence, we doing a composite numerical integration over subintervals of width h

Main difference: we can't parallelize because the integral $y' = f(t, y(t))$ depends on y itself.

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Backward Euler Method

Can we recover the backward Euler method with this approach? Yes, simply use the **right rectangle approximation** to the numerical integral:



$$f(t_{i+h}, y(t_{i+h}))$$

$$\int_{t_i}^{t_{i+h}} f(t, y(t)) dt$$

$$\approx h \cdot f(t_{i+h}, y(t_{i+h}))$$

$$y_{i+1} = y_i + h \cdot f(t_{i+1}, y_{i+1})$$

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The Trapezoid Rule

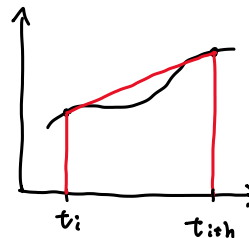
One more idea: what happens when we apply the trapezoid rule?

$$\int_{t_i}^{t_{i+h}} f(t, y(t)) dt \approx \frac{h}{2} [f(t_i, y(t_i)) + f(t_{i+h}, y(t_{i+h}))]$$

$$y_{i+1} = y_i + \frac{h}{2} [f(t_i, y_i) + f(t_{i+1}, y_{i+1})]$$

↳ Another implicit rule for solving IVP

we will relate this technique to the Runge-Kutta method next lecture



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Forward Euler for an Exponential Solution

Consider the following IVP:



This has a simple exact solution:

$$y' = \lambda y, \quad y(0) = y_0.$$

Assume $\lambda < 0$
 $\Rightarrow y(t)$ decrease

$$y(t) = e^{\lambda t} y_0.$$

Forward Euler (stepsize h)

$$|y_{i+1}| \leq |y_i|$$

$$y_{i+1} = y_i + h \cdot f(t_i, y_i)$$

$$= y_i + h \cdot \lambda y_i$$

$$= (1 + h\lambda) y_i = (1 + h\lambda)^2 y_{i-1}$$

$\vdots$

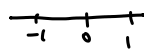
$$= (1 + h\lambda)^{i+1} y_0$$

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Stability for Forward Euler

$$y_{i+1} = (1 + h\lambda) y_i$$

For the method to be numerically stable, we require



$$|y_{i+1}| = |1 + h\lambda| \cdot |y_i| \leq |y_i|$$

$\Rightarrow |1 + h\lambda| \leq 1$, on expanding

$-1 \leq 1 + h\lambda \leq 1$ Assume $\lambda < 0$, and $h > 0 \Rightarrow$ RHS is satisfied.

$$-2 \leq h\lambda \leq 0$$

$$\Rightarrow h \leq \frac{2}{|\lambda|}$$

$$|-2| \geq |h| \cdot |\lambda|$$

$$h \leq \frac{2}{|\lambda|}$$

Therefore, large $|\lambda|$ can restrict our step size.

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Forward Euler Stability Example

Example 2: Consider this IVP for $\lambda = -10$:

$$y' = -10y, \quad y(0) = y_0.$$

Forward Euler is $y_{i+1} = y_i + h\lambda y_i = (1 - 10h)y_i$

For stability, we need $h \leq \frac{2}{10} = 0.2$

Try e.g. $h = 0.21$

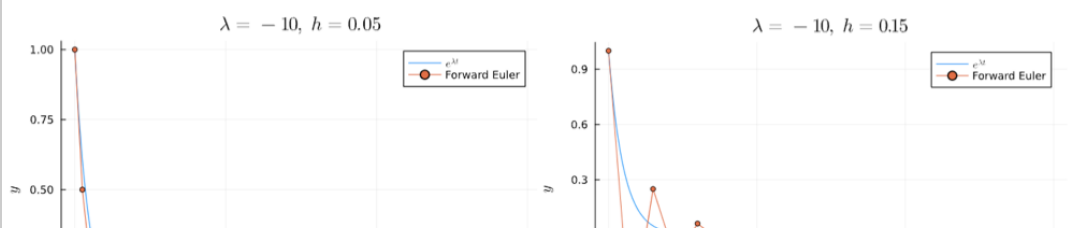
$$y_1 = (1 - 10 \times 0.21)y_0 = -1.1y_0$$

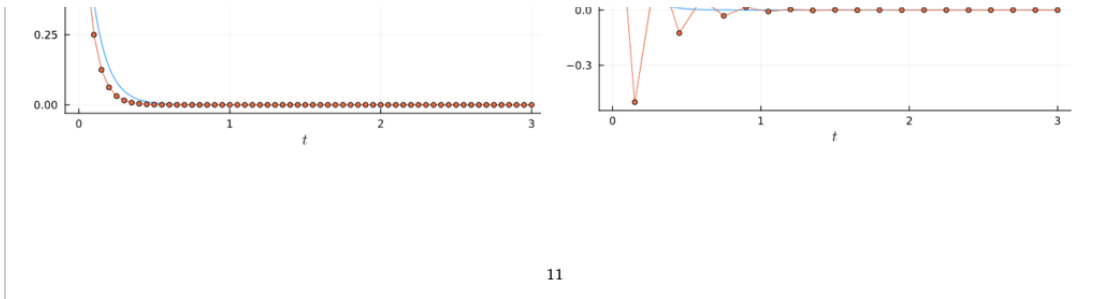
$$y_i = (-1.1)^i y_0$$

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Forward Euler Stability Visualized

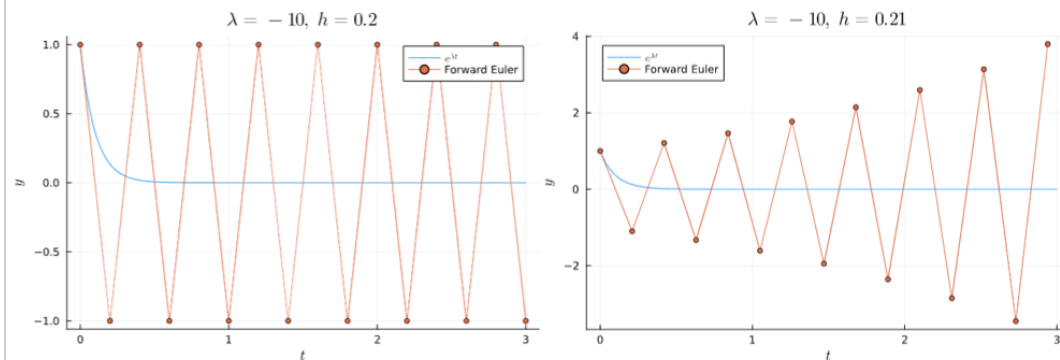
Example 2 cont.: The performance depends on the choice of h :





Forward Euler Stability Visualized

Example 2 cont.: Performance at and above the critical step size $h = 0.2$:



Stability for Backward Euler

Next consider the backward Euler method on the same family of IVPs:

$$y' = \lambda y, \quad y(0) = y_0, \quad |1 - h\lambda| > 1$$

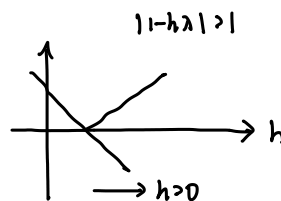
where $\lambda < 0$ once more.

$$y_{i+1} = y_i + h \cdot \lambda \cdot y_{i+1}$$

$$y_{i+1} = \frac{1}{1 - h\lambda} y_i \rightarrow \frac{1}{|1 - h\lambda|} < 1$$

$$|y_{i+1}| = \frac{1}{|1 - h\lambda|} |y_i| \leq |y_i|$$

Inherently stable for any $h > 0$



→ implicit methods are stable in general !!!

$\Rightarrow$ implicit methods are stable in general!

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Backward Euler Stability Example

Example 3: Consider this IVP for $\lambda = -10$ again:

$$y' = -10y, \quad y(0) = y_0,$$

but this time apply backward Euler.

$$y_{i+1} = \frac{y_i}{1 - \lambda h} = \frac{1}{1 + 10h}$$

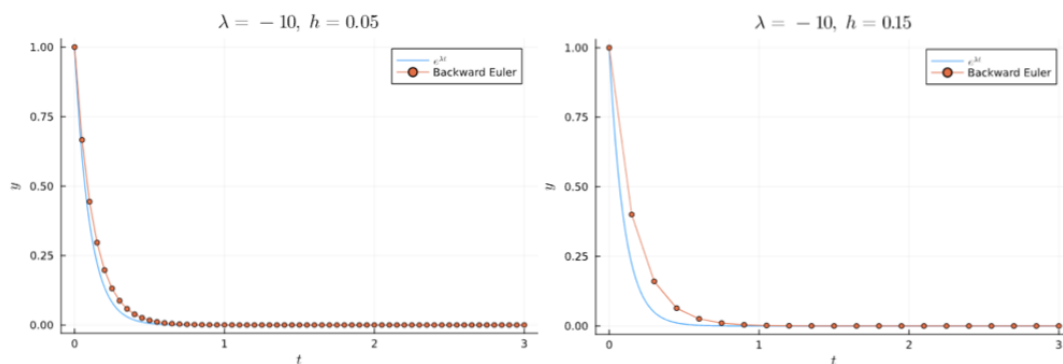
remains < 1 for any $h > 0$

$\Rightarrow$ stable even for large steps!

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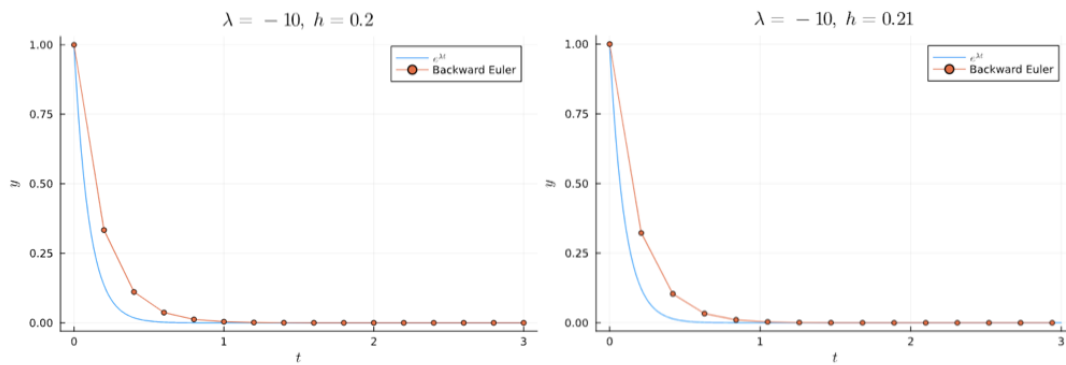
Backward Euler Stability Visualized

Example 3 cont.: The performance once again depends on the choice of h :



Backward Euler Stability Visualized

Example 3 cont.: Performance of the backward method at and above the forward method's critical step size $h = 0.2$:



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Local Truncation Error

LTE

Local truncation error is the amount by which the exact solution fails to satisfy the numerical method. For the forward Euler method, we have:

$$y_{i+1} = y_i + h \cdot f(t_i, y_i)$$

We can sub. in the actual solutions $y(t_{i+1}), y(t_i)$ and divide by h to define the LTE:

$$d_i \triangleq \frac{y(t_{i+1}) - y(t_i)}{h} - f(t_i, y(t_i))$$

$$= \frac{h}{2} y''(\xi_i), \quad \xi_i \in [t_i, t_{i+1}], \text{ where we used Taylor theorem}$$

$$y(t_{i+1}) = y(t_i) + h f(t_i, y(t_i)) + \frac{h^2}{2} y''(\xi_i)$$

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Order of a Numerical Solution Method

A method is of order q if q is the lowest positive integer such that for any sufficiently smooth exact solution $y(t)$ we have

$$\max_i |d_i| = O(h^q)$$

$\Rightarrow$ both forward (and backward) Euler methods are of order $q=1$ (first order)

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Global vs. Local Error

$y(t_i)$

Global error at t is the difference between the exact solution and the numerical

Global error at t_i is the difference between the exact solution and the numerical method's computed approximation:

$$e_i \triangleq y(t_i) - y_i, \quad i = 0, 1, \dots, n$$

Where $y(t_i)$ is once again the exact solution, y_i is our approx.

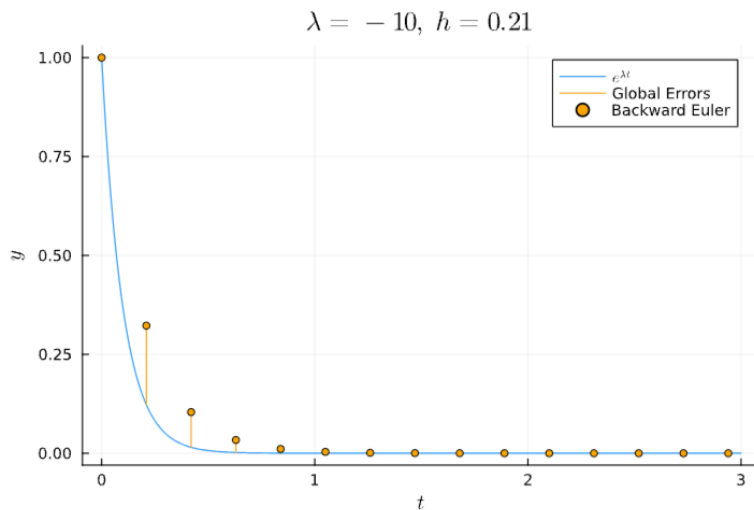
Consider $u'(t) = f(t, u)$, $u(t_{i-1}) = y_{i-1}$

I.e. this is an IVP from the previous step's computed solution. The local error is $\ell_i \triangleq u(t_i) - y_i$, where $u(t_i)$ is the exact solution to this new IVP over a single time step.

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Global Error Visualized

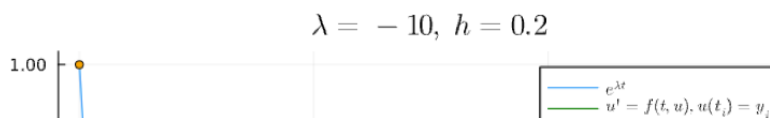
Let's look at the global error for backwards Euler on $y' = \lambda y$:

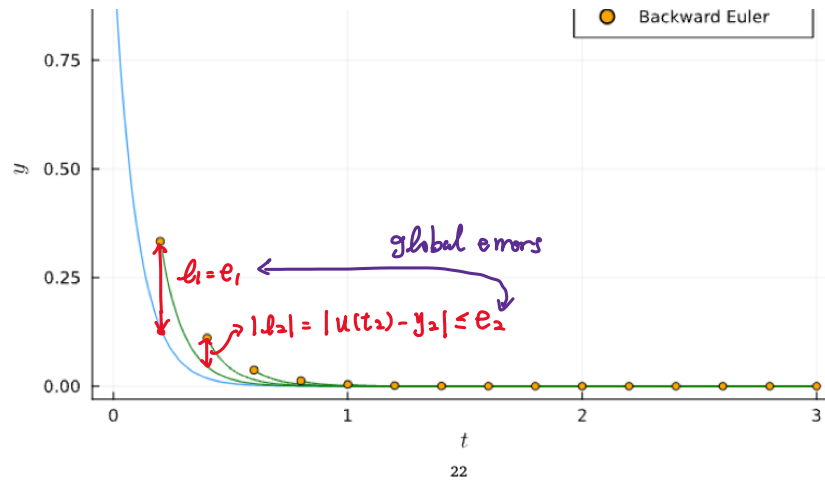


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Local Error Visualized

The local error is a little bit subtler:





Convergence

A numerical method is said to *converge* if the maximum global error goes to zero as the step size $h \rightarrow 0$:

$$\max_i e_i = \max_i [y(t_i) - y_i] \rightarrow 0 \text{ as } h \rightarrow 0$$

$\Rightarrow$ We can completely (at all time steps) with the step size h

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我们之前学习了forward和backward Euler method去数值计算一个微分方程，通过数值解（求点）的方法代替求解解析解。事实上，我们可以把Euler method和求积分结合起来，

1. 核心原理：微积分基本定理

- 看看右上角的手写笔记：

$$\int_a^b y'(t) dt = y(b) - y(a)$$

- 这告诉我们：如果我们想知道函数 y 下一步的值 $y(t+h)$ ，我们需要知道它当前的值 $y(t)$ 加上它导数的积分（即导数曲线下的面积）：

$$u(t+h) = u(t) + \int_t^{t+h} f(t, u(t)) dt$$

$$y(t) = y(t_0) + \int_{t_0}^t y'(t) dt$$

(因为 $y' = f$)

怎么计算这个积分部分就是这节课讨论的内容。

我们可以转换视角，现在我们知道Euler method不是由Taylor series得来的，而是通过不断计算在 $t_i \rightarrow t_{i+1}$ 上的小矩形然后累加。

注意其实我们是对导数 y' 积分，变相的就是对 $f(t, y)$ 积分。

然后我们想知道，什么时候Euler method是稳定的，比如说原函数是一个逐渐衰减的函数，那么我们算出来的函数是不是也是逐渐衰减的还是会发散。根据推导公式，我们可以整合为 $y_{i+1} = (1 + h\lambda)^{i+1} y_0$ ，所以只有当 $|1 + h\lambda| < 1$ 才会衰减。所以这让 h 必须足够小（对于Forward来说），

Backward永远不会发散。

LTE, local truncation error就是exact solution无法满足数值方法的程度。假设我们在这一步的起点是完全正确的（在真实曲线上），那么仅仅迈出这一步，我们会产生多大的误差？

叫local的原因就是我们只关注这一步的误差，Forward Euler的LTE是 $d_i = \frac{h}{2} y''(\xi_i)$, $\xi_i \in [t_i, t_{i+1}]$

那么global error就是所有误差的累积，每一步local error的累积。

所以convergence的定义就是，当 h 趋近于0的时候，最大的global error也趋近于0。

$y(t_{i+1})$ 是当自变量为 t_{i+1} 的真实值， y_{i+1} 是通过数值方法得到的近似值。

微分方程本身的稳定性是依赖于 λ ，或者 A ，对于 $y' = Ay/\lambda y$ ，需要 λ 本身 < 0 ，或者 A 不是PD/PSD，即eigenvalue是不是都小于0；这个稳定是指当自变量趋于 ∞ ， y 是不是也是趋于无穷。即使ODE不稳定也可以有解析解。

数值方法的稳定性是另一个东西，不要搞混。